

Table 3: Empirical Test

Empirical Test (Back Test, Forward Test, Paper Trading, Live Trading)	
Benchmark	Four baselines evaluated on the same 128-sample held-out 2022 World Cup eval set: (1) Base Llama 3.1 8B Instruct (no fine-tuning): 45.3% result accuracy, 0% exact score, 2.13 goal MAE. (2) Always Home Win: 45.3% result accuracy, 10.9% exact score, 1.11 goal MAE. (3) Random Weighted (40/35/25 home/away/draw): 35.9% result accuracy, 7.8% exact score. (4) Always Draw: 23.4% result accuracy. Football-LLM beats all baselines on result accuracy (52.3%) and dominates on exact score (29.7%).
Performance Assessment	Result accuracy: 52.3% (vs. 45.3% best baseline). Exact score match: 29.7% (vs. 0% base model). Goal MAE: 1.29 (40% improvement over base model's 2.13). Parse rate: 100%. Simulated ROI (flat bet, avg odds 2.3): +20.4% over 128 bets. Filtered ROI (top 50% confidence): +36.6%. Simulated max drawdown: 6.0%. Per-bet Sharpe ratio: 0.178. Winning probability: 52.3%. Trading frequency: ~4.4 bets/day during tournament. Average holding period: ~90 minutes (single match). Extendable to 2,000+ matches/year across leagues. Memorization test: Named teams 50.0% vs Anonymized 54.7% (model reasons from stats, not name recall).
Back Test 1	2022 FIFA World Cup, full tournament (Nov 20 to Dec 18, 2022). Group stage (48 matches) serves as the high-variance "bear" phase with frequent upsets (Saudi Arabia over Argentina, Japan over Germany, Cameroon over Brazil). Knockout stage (16 matches) serves as the lower-variance "bull" phase where favorites generally advance. Training data (2010 + 2014 + 2018 WCs) similarly spans both phases. The 52.3% accuracy and +20.4% simulated ROI reflect performance across both market conditions.
Back Test 2	The 2022 World Cup is the most recent completed FIFA World Cup tournament. For a nearest-period backtest on league data, the model can be evaluated on 2024-2025 domestic league seasons (Premier League, La Liga, Bundesliga, Serie A, Ligue 1) using the same player statistics pipeline and prompt format. The data

Back Test Scenario 1	infrastructure supports this: the API-Football scraper covers all major leagues. This extension provides 1,500+ additional out-of-sample matches and tests generalization beyond the tournament format.
	The 2022 World Cup contained the biggest upset in World Cup history: Saudi Arabia defeating Argentina 2-1 in the group stage. Argentina entered as FIFA #3, Copa America champions, and on a 36-match unbeaten run. Other "crisis" events include Japan beating Germany 2-1, Japan beating Spain 2-1, and Cameroon beating Brazil 1-0. These matches are the sports equivalent of black swan events. The model's reasoning chain on these matches often identifies relevant defensive signals (e.g., high tackles per 90, duel win %) even when the overall prediction still favors the stronger team.
Back Test Scenario 2	Knockout stage matches (Round of 16 through the Final, 16 matches) as a stress test. These feature the strongest remaining teams with the smallest margins. Many are decided by a single goal or penalties. The model's calibration on closely-matched teams is tested here, where small differences in player quality and tactical matchups are the primary differentiators. This is the equivalent of a low-volatility, high-stakes trading environment.
Paper Trading (Forward Test)	Forward test plan: run the model on 2025-2026 domestic league fixtures and World Cup 2026 qualifying matches. For each match, generate a prediction with reasoning 24-48 hours before kickoff, logged alongside bookmaker odds at prediction time. After the match, compare against actual results and compute simulated P&L. No real bets placed. This phase calibrates the edge threshold (T_edge) on a larger, more diverse sample and validates out-of-tournament generalization. In sports betting, frictions are minimal (no slippage or partial fills), but odds can shift between prediction and kickoff. Paper trading measures this odds drift.
Live Trading	Not yet implemented. Planned after paper trading produces sufficient out-of-sample validation. Live execution requires: sportsbook API integration for automated bet placement, real-time odds monitoring to confirm edge exists at execution time, and an automated bankroll management system enforcing Kelly sizing, per-match exposure caps, and drawdown circuit breaker. Target: World Cup 2026 (June-July 2026) as the first live deployment, with domestic league betting as the extended application.
Capacity Test	Sports betting markets have lower capacity than financial markets. Major sportsbooks can absorb \$50K-\$100K per outcome on World Cup and top league matches before significantly moving the line. The strategy targets bankrolls under \$50K where individual bets (1-5% of bankroll = \$500-\$2,500) are well within absorption limits. For larger scale, diversification across multiple sportsbooks or

	betting exchanges (e.g., Betfair) would be needed to avoid adverse line movement.
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